

## Practical 6

### Implementing Polymorphism in Java

1) Write a Java program to create a class Shape with methods getArea() and getPerimeter(). Create three subclasses: Circle, Rectangle, and Triangle. Override the getArea() and getPerimeter() methods in each subclass to calculate and return the area and perimeter of the respective shapes. [ Hint for triangle area =  $\sqrt{s(s-a)(s-b)(s-c)}$  where  $s = (a+b+c)/2$  ]

```
->class Shape {
    public void getArea() {}
    public void perimeter(){}
}

class Circle extends Shape {
    double r;
    Circle(double r)
    {
        this.r=r;
    }
    public void getArea()
    {
        double re =3.14*(r*r);
        System.out.println("The area of Circle :"+re);
    }
    public void perimeter()
    {
        System.out.println("Perimeter of Circle : "+(2* Math.PI *r));
    }
}
```

```

}
class Rectangle extends Shape {
    double le,wi;
    Rectangle(double le,double wi)
    {
        this.le=le;
        this.wi=wi;
    }
    public void getArea()
    {
        double re=le*wi;
        System.out.println("The area of Rectangle :"+re);

    }
    public void perimeter()
    {
        System.out.println("Perimeter of Rectangle :"+(2* (le+ wi)));
    }
}
class Triangle extends Shape {
    double a,b,c;
    Triangle(double a,double b,double c)
    {
        this.a=a;
        this.b=b;
        this.c=c;
    }
    public void getArea()

```

```

{
    double re=(a+b+c)/2;
    double finalre=Math.sqrt(re * (re -a)*(re-b)*(re-c));
    System.out.println("The area of Triangle :"+finalre);
}
public void perimeter()
{
    System.out.println("Perimeter of Triangle :"+(a+b+c));
}
}

public class Q1 {
    public static void main(String[] args) {
        Circle c=new Circle(10);
        Rectangle r=new Rectangle(10,20 );
        Triangle tr=new Triangle(20, 30, 40);
        c.getArea();
        c.perimeter();
        r.getArea();
        r.perimeter();
        tr.getArea();
        tr.perimeter();
    }
}

```

=> Output:-

The area of Circle :314.0

Perimeter of Circle:62.83185307179586

The area of Rectangle :200.0

Perimeter of Rectangle:60.0

The area of Triangle :298.1423949999899

Perimeter of Triangle:90.0

2) Write a Java program to create a class Employee with a method called calculateSalary(). Create two subclasses Manager and Programmer. In each subclass, override the calculateSalary() method to calculate and return the salary based on their specific roles.

```
->class Employee{
    double salary;
    public double calculateSalary()
    {
        return salary;
    }
}

class Manager extends Employee {
    double basesalary;
    double overtime;
    double rate=1000;
    Manager(double basesalary,double overtime)
    {
        this.basesalary =basesalary;
        this.overtime=overtime;
    }
    public double calculateSalary()
    {
        double finalre= basesalary + (overtime * rate);
        return finalre;
    }
}
```

```

    }
}
class Programmer extends Employee {
    double basesalary;
    double overtime;
    double rate=500;
    Programmer(double basesalary,double overtime)
    {
        this.basesalary =basesalary;
        this.overtime=overtime;
    }
    public double calculateSalary()
    {
        double finalre= basesalary + (overtime * rate);
        return finalre;
    }
}
public class Q2 {
    public static void main(String[] args) {
        Manager mg=new Manager(50000, 20);
        System.out.println(mg.calculateSalary());
        Programmer pg=new Programmer(30000, 20);
        System.out.println(pg.calculateSalary());
    }
}

```

=>Output:-

70000.0

40000.0

3) Write a Java program that overloads a method display() to: Display a single string, Display a string and an integer, Display two strings

```
->public class Q3 {  
    public void display(String s){  
        System.out.println(s);  
    }  
    public void display(String s1,String s2){  
        System.out.println(s1);  
        System.out.println(s2);  
    }  
    public void display(String s,int a){  
        System.out.println(s);  
        System.out.println(a);  
    }  
    public static void main(String[] args) {  
        Q3 s=new Q3();  
        s.display("hill");  
        System.out.println();  
        s.display("hill", 20);  
        System.out.println();  
        s.display("Hello","world");  
    }  
}
```

=>Output:-

hill

hill

20

Hello

World

4) Create a class TemperatureConverter that overloads a method convert() to:  
Convert Celcius To Farenheit And Convert Farenheit To Celcius.

```
->class TemperatureConverter{
    double temperature;
    public double convert(double Celsius)
    {
        return (Celsius * 9/5) +32;
    }
    public double convert( int Fahrenheit)
    {
        return (Fahrenheit - 32) * 5.0 /9.0;
    }
}

public class Q4 {
    public static void main(String[] args) {
        TemperatureConverter te=new TemperatureConverter();
        System.out.println( te.convert(30.00));
        System.out.println(te.convert(50));

    }
}
```

```
}
```

=>Output:-

86.0

10.0

5) Create a class Employee that overloads a method calculateSalary() for: Fixed salary, Salary with bonus, Salary with bonus and overtime.

```
-> class Employee {  
    public void calculateSalary(double fixedsal)  
    {  
        System.out.println("Fixed salary :"+fixedsal);  
    }  
    public void calculateSalary(double fixedsal,double bonus)  
    {  
        double finalre =fixedsal +bonus;  
        System.out.println("Fixed salary with bonus :"+finalre);  
    }  
    public void calculateSalary(double fixedsal,double bonus,double overtime)  
    {  
        double finalre =fixedsal +bonus + (overtime * 1000);  
        System.out.println("Fixed salary with bonus and overtime :"+finalre);  
    }  
}  
  
public class Q5 {  
    public static void main(String[] args) {  
        Employee e1=new Employee();
```



```

        e1.calculateSalary(20000.0);
        e1.calculateSalary(20000.0,1200.0);
        e1.calculateSalary(20000.0,1200.0,3.0);
    }
}

```

=>Output:-

Fixed salary :20000.0

Fixed salary with bonus :21200.0

Fixed salary with bonus and overtime :24600.0

6) Write a program that demonstrates method overloading with varargs:  
 multiply (float... numbers) multiply(double... numbers).

```

->class Calculator {
    public void multiply(float... numbers) {
        float result = 0.0f;
        for (float num : numbers) {
            result *= num;
        }
        System.out.println("Multiplication result " + result);
    }
    public void multiply(double... numbers) {
        double result = 1.0;
        for (double num : numbers) {
            result *= num;
        }
        System.out.println("Multiplication result " + result);
    }
}

```

```
    }  
}  
public class Q6 {  
    public static void main(String[] args) {  
        Calculator calc = new Calculator();  
        calc.multiply(12.0, 30.0, 3.0);  
        calc.multiply(2.0, 3.0, 4.0, 5.0);  
    }  
}
```

=>Output:-

Multiplication result 1080.0

Multiplication result 120.0